Measure sound speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Analyze sound travel measurements and their limitations.

SUCCESS CRITERIA - I can...

1. Define one-way versus round-trip paths.
2. Calculate speed from distance and time.
3. Use repeated measurements and discuss timing uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Three instantaneous displacements are **+3 mm**, **-5 mm** and **-7 mm**. **Find their sum** and **explain why one displacement value** does not find total wave energy.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: v = d/Δt; v = **2** d/Δt for an echo.
Match the distance to one-way or round-trip travel. Use moderate sound levels and **record actual data separately**.

Worked reasoning: A sound echo returns **0.4 s** after a pulse. At **340 m/s**, round-trip distance is **136 m**, so the reflector is **68 m** away.

Your turn: Draw outgoing and returning paths and **mark where the factor of 2** enters.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A pulse travels between two sensors separated by **1.70 m**. Delay is **0.0050 s**. Is this one-way or echo distance? **Find speed**.



Marzano check - circle: ① ② ③

04 Apply the physics

A microphone detects an echo **0.0200 s** after a pulse hits the air. The reflecting wall is **3.40 m** away. **Find sound speed**. Treat distance as exact and timing as **0.0200 ± 0.0005 s**; **calculate the lowest** and highest possible speeds. **Label the round trip**.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Plan a table for three trials. Include distance, elapsed time, method (one-way or echo), speed and one possible systematic error.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A sphere’s radius becomes **4** times as large at fixed density. Derive the area-to-mass multiplier. **Explain why kinetic energy per** mass alone does not predict survival during entry.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

07 Exit check

Two sensors **0.85 m** apart record a wave’s travel time as **0.0025 s**. **Find the wave speed**. A timing fault adds **0.001 s** to the recorded interval. **Calculate the speed this faulty** measurement would give, and **explain the direction of the** error. **Label the one-way path**.



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A microphone pair separated **1.5 m** measures travel-time delay **0.0044 s**. **Find speed**. If delay is uncertain by±**0.0002 s**, find worst-case speed bounds.



Marzano check - circle: ① ② ③